

SHAKTHI S

✉ shakthi20siva@gmail.com  linkedin.com/in/shakthi-s-030440249  github.com/shakthi-20

EDUCATION

Bachelor of Technology, Electrical and Electronics Engineering

Amrita Vishwa Vidyapeetham | CGPA: 8.16

Oct 2022 – Mar 2026

Coimbatore, India

Relevant Coursework : Signals and Systems, Digital Signal Processing, IoT, Fundamentals of Soft Computing, AI and Edge Computing, Machine Learning, Deep Learning for Visual Computing and Cybersecurity.

SKILLS

Languages: Python, C, HTML5, CSS

Libraries & Frameworks: Pandas, NumPy, TensorFlow, TensorFlow Lite, PyTorch, Keras, scikit-learn, Flask, SQLAlchemy

Databases & Cloud: MySQL, Firebase

Tools: Git, GitHub, VS Code, MATLAB

Embedded & Hardware: ESP32, Arduino, Raspberry Pi

Operating Systems: Kali Linux, Raspberry Pi OS, Linux

Domains: IoT Systems, Edge AI, REST API Integration, Network Protocols (MQTT)

PROFESSIONAL EXPERIENCE

Internship Trainee, Satish Dhawan Space Centre SHAR, ISRO

May 2025 – Jun 2025

Sriharikota, India

- Analysed Multi-Object Tracking Radar (MOTR) and antenna arrays supporting real-time signal transmission for rocket launches.
- Authored technical documentation, system workflow diagrams, and safety protocols for mission-critical launch operations.

In-plant Trainee, Kudankulam Nuclear Power Project, NPCIL

Dec 2024

Kudankulam, India

- Performed diagnostics on high-voltage systems (transformers, circuit breakers); prepared maintenance & compliance reports under nuclear safety standards.

PROJECTS

Detection of False Data Injection Attacks in Smart Micro-Grid Using Edge AI

Implemented a ML-based cyber attack detection using ESP32–Raspberry Pi MQTT communication, with Firebase-based cloud logging and real-time monitoring dashboards for a three bus system.

VisionSole - Smart Assistant for the Visually Impaired

Developed a wearable assistive system using ESP32-CAM and sensors for real-time obstacle detection, integrated with an Android app using Google Maps API for voice navigation and emergency alerts.

Secure IoT Node for OTP-Based Access and Tamper Detection

Built a dual-ESP32 smart locker using OTP authentication and tamper detection, with MQTT-based node communication and real-time alerts via Blynk Cloud.

Line Following Robot

Designed and built a microcontroller-free line-following robot using LM358 op-amp and L293D IC.

Simulation and CAN-Based Implementation of an Anti-lock Braking System

Developed and simulated ABS in MATLAB/Simulink using Bang-Bang and Fuzzy Logic controllers for braking performance. Implemented CAN communication in Python for real-time data transfer, improving safety and stability.

IoT-Based Plant Leaf Health Monitoring

Trained & deployed CNN (73.5% accuracy) for leaf disease detection on ESP32 via TFLite; mobile dashboard alerts cut manual inspection ~50%.

PUBLICATIONS

Breast Cancer Screening from Thermal Images: CNN-Based Analysis with Augmented Datasets and Web App

Deployment - IEEE INSPECT 2025

Predictive Modeling of Video Streaming Effectiveness Using Artificial Neural Network and Sentiment Analysis - IEEE

ICERECT 2025

Spectrum-Agnostic Space Debris Detection with Autoencoders - IEEE ICAUC 2026

CERTIFICATIONS

AI Agents 101: Building AI Agents with MCP and Open-Source Interface - **AMD AI Academy** | Foundation in Data Science and Applications - **IIT Madras** | Artificial Intelligence and Edge Computing - **L&T Edutech** | MATLAB OnRamp - **MathWorks**

EXTRA-CURRICULARS

Volunteer - **National Service Scheme (NSS)** | Electronics Engineer - **Amrita Robotics Club** | Social Media Team Designer - **Amrita Animal Welfare Society**